

Small Office Network Lab: Project Brief

Oct 6, 2026 · @blake

Scenario and constraints

This project builds a segmented, redundant small-office network on two physical Cisco Catalyst Layer 3 switches, then validates it through planned failure tests.

How this brief was created: I asked Claude, an AI assistant made by Anthropic, to write these requirements so I could simulate a client project on real hardware. Claude provided the goals and acceptance criteria only. I designed the implementation, wrote every configuration, ran every test, and wrote the documentation myself.

The simulated client is a small office that needs:

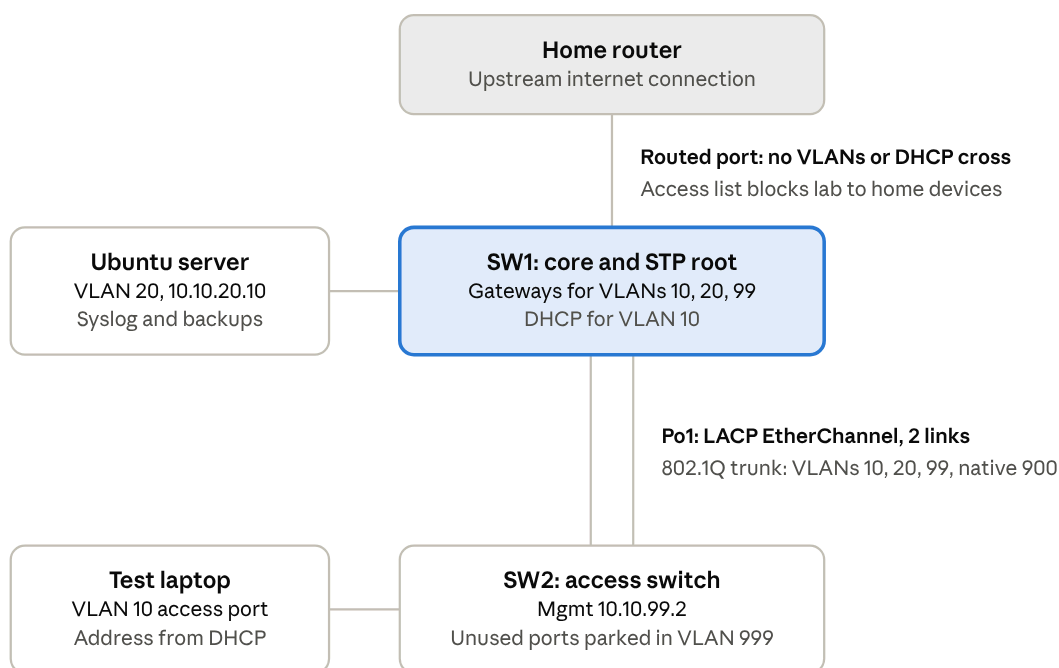
- Users, servers and network management on separate networks
- An inter-switch uplink that survives a single cable failure
- Internet access through an existing upstream router, without the office network extending onto or interfering with the upstream network
- Secure remote management, centralized logging and configuration backups

Constraints: two Catalyst Layer 3 switches, one Ubuntu server, one test laptop, and a consumer home router as the only path to the internet. All work is done on physical hardware, not in a simulator.

Topology and addressing

SW1 is the core switch, spanning tree root and gateway for every lab VLAN, and it is the only device that connects to the upstream home router.

Lab traffic reaches the internet through one routed port on SW1



lab topology · home router uplink, 2 switches, 2 test devices

The uplink to the home router is a routed port, so it carries no lab VLANs and no lab DHCP traffic. The two inter-switch links act as one logical link, so losing either cable should not interrupt traffic.

VLAN	Name	Subnet	Gateway	Purpose
10	USERS	10.10.10.0/24	10.10.10.1	User devices, addressed by DHCP
20	SERVERS	10.10.20.0/24	10.10.20.1	Ubuntu server, static 10.10.20.10
99	MGMT	10.10.99.0/24	10.10.99.1	Switch management: SW1 .1, SW2 .2
900	NATIVE	none	none	Trunk native VLAN, carries no hosts
999	PARKING	none	none	Unused ports, shut down
Uplink	none	Home network	Home router	SW1 routed port to the home router

Requirements

Each requirement is complete only when its acceptance criteria are met and the verification output is saved in the repo.

#	Requirement	Accepted when	Status
1	Clean baseline	Both switches are inventoried (model, software, capabilities) and reset to a blank configuration	Not started
2	Secure management	Hostnames, encrypted privileged password, local user login and banner are set; remote access is SSH version 2 only and Telnet is refused	Not started
3	Segmentation	VLANs 10, 20 and 99 exist on both switches; every unused port is in VLAN 999 and shut down	Not started
4	Redundant uplink	Two links form one LACP EtherChannel; the 802.1Q trunk carries only lab VLANs, uses VLAN 900 as native and does not negotiate trunking	Not started
5	Spanning tree	Rapid PVST+ runs on both switches; SW1 is root for every lab VLAN and SW2 is backup; edge ports use PortFast and BPDU Guard	Not started
6	Inter-VLAN routing	SW1 routes between VLANs; a VLAN 10 device reaches the server in VLAN 20	Not started
7	DHCP	SW1 assigns VLAN 10 addresses automatically, reserving .1 through .20, with the correct gateway	Not started
8	Logging and backups	The Ubuntu server receives timestamped syslog from both switches and stores both running configurations	Not started

#	Requirement	Accepted when	Status
9	Internet access	Lab devices reach the internet through the home router over a routed port; lab VLANs and DHCP stay off the home network, and an access list blocks lab devices from home devices	Not started
10	Access-layer security (optional)	The user port allows one learned device, and DHCP snooping protects VLAN 10	Not started

Validation and failure testing

Each test runs during a continuous ping from the test laptop to the server, and the observed result is recorded alongside the pass criteria.

Test	What it proves	Pass criteria	Result
Single uplink failure	EtherChannel redundancy	Traffic continues over the remaining link and the bundle stays up	
Both uplinks fail, then recover	Recovery behavior	Traffic resumes once links are restored; recovery time is recorded	
BPDU received on an edge port	Loop protection	The port is error-disabled and the event is logged	
Native VLAN mismatch	Trunk misconfiguration detection	The mismatch is detected and logged by the switches	
Root bridge change	Spanning tree control	The root moves to SW2 as configured and returns when priorities are restored	
Device in the wrong access VLAN	Segmentation	The device lands outside VLAN 10 and the cause is identified from switch output	

Test	What it proves	Pass criteria	Result
Lab device attempts to reach a home device	Access list enforcement	The connection is blocked while internet access still works	
Second device on a secured port (optional)	Port security	The violation is detected and logged	

Deliverables

The GitHub repository contains:

- A README with design decisions and the reasoning behind each one
- A topology diagram and addressing plan
- Sanitized running configurations for both switches, with all credentials removed
- Verification output for each requirement
- Failure test results
- A troubleshooting log of problems hit during the build, how each was diagnosed, and the fix

Skills demonstrated

Area	Requirements	Skills
Switching	3, 4, 5	VLANs, 802.1Q trunking, LACP EtherChannel, Rapid PVST+, PortFast, BPDU Guard
Routing	6, 9	SVIs, inter-VLAN routing, routed ports, static default routing
Network services	7, 8	DHCP, syslog, TFTP configuration backups, Linux server administration
Security	2, 9, 10	SSH hardening, access lists, port security, DHCP snooping
Operations	All	Failure testing, structured troubleshooting, technical documentation